

Theory of Automata

Assignment # 3

Deadline: 5 December 2025

Instructions:

1. Attempt all questions.
2. Questions must be attempted on paper (A4 sheets or loose sheets) 3. Submit the assignment on time. Only physical submissions are accepted.

Questions 1:

Design PDA for given Context Free Language.

- I. To design a context free grammar which can produce balance parenthesis consider in alphabet set we have $\{ (,) \}$
- II. Design a CFG and PDA for even and odd palindrome
 - a. Even palindrome 001 100
 - b. Odd palindrome 00100
- III. Design a CFG and PDA for language $\{01(1100)^n 110 (10)^n \mid n \geq 0\}$
- IV. Design a CFG and PDA for language $\{0^n 1^m \mid n \neq m\}$
- V. Design a CFG and PDA for language $\{ (0^n 1^n 2^m 3^m \mid n \geq 1, m \geq 1) \text{ or } (0^n 1^m 2^m 3^n \mid n \geq 1, m \geq 1) \}$
- VI. Design a CFG and PDA for language $\{ (0^i 1^j 2^k \mid k \leq i, k \leq j) \}$
- VII. Design a CFG and PDA for language $\{ (0^i 1^j 2^k \mid i=j, j=k) \}$
- VIII. Design a CFG and PDA for language $\{ (0^n 1^m 2^{2m} \mid n, m \geq 0) \}$
- IX. Design a CFG and PDA for language $\{ (0^n 1^m \mid n \leq m+3) \}$
- X. Design a CFG and PDA for language $\{ (0^n 1^m \mid 2^n \leq m \leq 3^n) \}$

Questions 2:

Scenario-Based Question:

You are developing a **syntax validation module** for a simple **programming language interpreter** that checks **nested structures** like parentheses, curly braces, and square brackets.

To model this system, you are required to **design a Pushdown Automaton (PDA)** that verifies the correctness of **balanced and properly nested symbols** in program code.

Description of the Problem:

A valid program string contains balanced pairs of $()$, $\{\}$, and $[]$ such that:

1. Every opening symbol has a corresponding closing symbol of the same type.
 2. Symbols are **properly nested** — e.g., $\{ [()] \}$ is valid but $\{ () \}$ is not.
 3. The input alphabet $\Sigma = \{ (,), \{, \}, [,] \}$.
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Tasks:

- a) Define the **formal 7-tuple** $(Q, \Sigma, \Gamma, \delta, q_0, Z_0, F)$ of the PDA that accepts valid nested symbol strings.
- b) Explain the **working principle** of your PDA (push and pop behavior).
- c) Draw the **state transition diagram** of your PDA.
- d) Trace the **step-by-step stack operations** for the input string $\{ [()] \}$.
- e) Write an **equivalent Context-Free Grammar (CFG)** that generates the same language accepted by your PDA.

Questions 3: Turing Machines

Create Turing Machines for the following language:

- i. $L_1 = \{0^n 1^m 0^n 1^m \mid n, m \geq 1\}$.
- ii. $L_2 = \{a^{3n} b^n c^{2n} \mid n \geq 2\}$.
- iii. $L_2 = \{a^{n+m} b^n c^m \mid n \geq 0\}$.
- iv. $L_3 = \{0^n \mid n \text{ is a power of } 2\}$

Create Turing Machines for the following function:

- v.
$$f(x, y) = \begin{cases} x+y & x < y \\ \text{"zero"} & x \geq y \end{cases}$$
- vi.
$$f(x, y) = \begin{cases} x+y & x < y \\ x-y & x > y \end{cases} \text{ else write "equal" on the tape}$$

if $x = y$.
- vii. 4^3 .
- viii. Design a TM to **divide two unary numbers** and output quotient + remainder.
- ix. Design a TM to **compare two unary numbers** and output the larger one
- x. Modify the palindrome TM to accept only **odd-length** palindromes.